INCH-POUND

MIL-F-63446B (AR)
AMENDMENT 1
13 January 2000

MILITARY SPECIFICATION

FUZES, GENERAL PURPOSE GRENADE: M230 AND M231 (PRACTICE)

This amendment forms a part of Military Specification MIL-F-63446B, dated 12 September1994, and is approved for use by the US Army Armament, Research, Development, and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1.1: Delete " MIL-A-70625 Automated Acceptance Inspection Equipment for Non-Electronic Components, Design, Testing and Approval, Of".

PAGE 6

List of Drawings:

Delete "Plate Assembly, Barrier	9345166	50" and substitute
"Plate Assembly, Barrier	9345166	25".

PAGE 16

TABLE I. First article inspection:

Delete note (a) in its entirety.

Delete "4.5.15(a)/4.5.16(a)" and substitute "4.5.15/4.5.16" in three (3) places.

Add "Ball to wall inspection	239 min(f)	3.4	Gage".

AMSC N/A FSC 1340
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TABLE I. First article inspection:

Delete note a in its entirety.

Delete "4.5.15(a)/4.5.16(a)" and substitute "4.5.15/4.5.16" in two (2) places.

Delete "4.5.15(a)/4.5.16(c)" and substitute "4.5.15/4.5.16".

Add "Ball to wall inspection 239 min(f) 3.4 Gage".

PAGE 20

4.4.2.1: Delete in its entirety and substitute included PAGE 20.

PAGE 21

4.4.2.1: Delete in its entirety and substitute included PAGE 21.

PAGE 22

4.4.2.2: Delete in its entirety and substitute included PAGE 22.

PAGE 23

4.4.2.2: Delete in its entirety and substitute included PAGE 23.

PAGE 24

4.4.2.2: Delete in its entirety.

PAGE 30

4.4.2.8:

Major 106: Delete "True position of true diameter to major body diameter" and substitute "True position of tip diameter to major body diameter".

Minor 204: Delete "Evidence of poor workmanship	Level IV	3.7	Visual"
and substitute "Edge between major diameter and shank	Level IV	3.4	Gage"

Minor 205: Add "Evidence of poor workmanship Level IV 3.7 Visual"

PAGE 31

4.4.2.9:

Add new major 109 and 110 as follows:

"109	Length of coating	Level III	3.4	Gage"
"110	Color of cured coating	Level III	3.4	Visual"

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Slider	1 OF 2	9333807

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 9333825/9345168 / INSPECTION METHOD REFERENCE
Critical	None defined.			
Special (b)				
1	Presence of red lacquer on bore rider	100%	3.4	AAIE
Major				
101	Height of cage	Level III (a)	3.4	Gage
102	Diameter of trigger cage hole	Level III (a)	3.4	Gage
103	True position of trigger cage hole	Level III (a)	3.4	Gage
104	Perpendicularity of cage	Level III (a)	3.4	Gage
105	Width of slider body	Level III (a)	3.4	Gage
106	Thickness of slider body ends	Level III (a)	3.4	Gage
107	Width of bore rider	Level III (a)	3.4	Gage
108	Length of bore rider to notch	Level III (a)	3.4	Gage
109	Depth of detonator hole	Level III (a)	3.4	Gage
110	True position of detonator counterbore	Level III (a)	3.4	Gage
111	True position of arming pin hole	Level III (a)	3.4	Gage
112	True position of safety pin hole	Level III (a)	3.4	Gage
113	Distance from slider side to end of rack	Level III (a)	3.4	Gage
114	Gear teeth parameters	Level III (a)	3.4	Gage
115	Straightness of slider	Level III (a)	3.4	Gage
116	Minimum coverage of red lacquer on bore rider	Level III (a)	3.4	Gage

NOTES:
(a). See paragraph 3.4.1 for applicability of reduced inspection.
(b). See MIL-A-48078 for definition.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.1	Slider	SHEET 2 OF 2	9333807

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 9333825/9345168 / INSPECTION METHOD REFERENCE
Minor				
201	Thickness of cage	Level IV (a)	3.4	Gage
202	Thickness of slider body between ends	Level IV (a)	3.4	Gage
203	Thickness of bore rider	Level IV (a)	3.4	Gage
204	Length of bore rider from arming pin hole	Level IV (a)	3.4	Gage
205	Diameter of detonator hole	Level IV (a)	3.4	Gage
206	Location of detonator slot	Level IV (a)	3.4	Gage
207	Thickness of detonator slot	Level IV (a)	3.4	Gage
208	Diameter of arming pin hole	Level IV (a)	3.4	Gage
209	Diameter of safety pin hole	Level IV (a)	3.4	Gage
210	Maximum coverage of red lacquer on bore rider	Level IV (a)	3.4	Gage
211	Evidence of poor workmanship	Level IV (a)	3.7	Visual

NOTES:
(a). See paragraph 3.4.1 for applicability of reduced inspection.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Body, Fuze	1 OF 2	9333824

NEXT HIGHER ASSEMBLY: 9347183/9347185

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Porosity of firing pin hole	100%	3.4.3	4.5.1
102	True position of firing pin cap counterbore	Level III	3.4.2	Gage
103	Diameter of firing pin bore	Level III	3.4.2	Gage
104	True position of firing pin bore	Level III	3.4.2	Gage
105	Width of slider slot	Level III (a)	3.4.2	Gage
106	Depth of slider slot	Level III (a)	3.4.2	Gage
107	Width of bore rider slot end	Level III (a)	3.4.2	Gage
108	Length of bore rider slot	Level III (a)	3.4.2	Gage
109	Depth of cage pocket	Level III (a)	3.4.2	Gage
110	Diameter of detent ball hole	Level III	3.4.2	Gage
111	True position of detent ball hole	Level III	3.4.2	Gage
112	True position of pivot screw hole	Level III	3.4.2	Gage
113	Location of pivot screw surface from front of slider slot	Level III	3.4.2	Gage
114	True position of center hole to slider slot	Level III	3.4.2	Gage
115	Diameter of arming pin counterbore	Level III	3.4.2	Gage
116	True position of arming pin counterbore to center hole	Level III	3.4.2	Gage
117	True position of timer holes	Level III	3.4.2	Gage
118	Diameter of drive screw hole	Level III	3.4.2	Gage
119	Diameter of fuze body	Level III (a)	3.4.2	Gage

NOTES:
(a). Reduced sampling for cast dimensions see 4.4.2 c.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Body, Fuze	2 OF 2	9333824

NEXT HIGHER ASSEMBLY: 9347183/9347185

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Major				
201	Diameter of firing pin cap counterbore	Level IV	3.4.2	Gage
202	Distance of firing pin cap counterbore to side of slider slot	Level IV	3.4.2	Gage
203	Width of key	Level IV (a)	3.4.2	Gage
204	Diameter of hub	Level IV (a)	3.4.2	Gage
205	Overall height	Level IV (a)	3.4.2	Gage
206	Height of intermediate diameter	Level IV (a)	3.4.2	Gage
207	Diameter of safety pin hole	Level IV	3.4.2	Gage
208	True position of safety pin hole	Level IV	3.4.2	Gage
209	Depth of bore rider slot	Level IV (a)	3.4.2	Gage
210	Diameter of render safe hole	Level IV	3.4.2	Gage
211	Depth of relief surface	Level IV (a)	3.4.2	Gage
212	Diameter of center hole	Level IV	3.4.2	Gage
213	Diameter of anti-twist pin holes	Level IV	3.4.2	Gage
214	True position of anti-twist pin holes	Level IV	3.4.2	Gage
215	Diameter of timer holes	Level IV	3.4.2	Gage
216	Height of slider guide rails	Level IV (a)	3.4.2	Gage
217	Thickness of flange	Level IV (a)	3.4.2	Gage
218	Chamfer of flange	Level IV (a)	3.4.2	Gage
219	Orientation of notches (First Article Only)	Level IV	3.4.2	Gage
220	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:
(a). Reduced sampling for cast dimensions see 4.4.2 c.

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4.4.2.9:

In the notes block, add notes 1 and 2 as follows:

"1. All characteristics inspected before coating."
"2. Major 101 and 104 measured before and after coating."

PAGE 50

4.4.2.28:

Major 103: Delete "Concentricity of minor diameter" and substitute "Run out of minor diameter".

PAGE 51

4.4.2.29:

Major 102: Delete "Visual" and substitute "Gage or SME".

PAGE 52

4.4.2.30: Delete in its entirety and substitute included PAGE 52.

PAGE 54

4.4.2.32:

Add new minor 203 as follows:

"203 Nose diameter Level IV 3.4 Gage"

PAGE 63

4.4.2.41:

Minor 201: Delete " 201 Evidence of poor workmanship" and substitute "203 Evidence of poor workmanship".

PAGE 69

4.4.2.47:

Delete "AAIE (g)" and substitute "AAIE" in two (2) places.

Delete "4.5.15(b)/4.5.16(b)" and substitute "4.5.15/4.5.16" in three (3) places.

Special (a): Add "12 Ball to wall 100% 3.4 Gage".

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.30	Body, Firing Pin	1 OF 1	9333816

NEXT HIGHER ASSEMBLY: 9345130

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Large diameter	Level III	3.4	Gage
102	True position of large diameter to nose	Level III	3.4	Gage
103	Straight pin hole diameter	Level III	3.4	Gage
104	True position of hole to nose	Level III	3.4	Gage
105	Detent ball angle	Level III	3.4	Gage
106	Surface finish of detent ball angle	Level III	3.4	Visual
Minor				
201	Diameter of tail	Level IV	3.4	Gage
202	True position of tail to nose	Level IV	3.4	Gage
203	Maximum radius on head	Level IV	3.4	Gage
204	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

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4.4.2.47:

Delete note (b) in its entirety.

Delete note (g) in its entirety.

PAGE 71

4.4.2.48:

Delete “AAIE (g)” and substitute “AAIE” in two (2) places.

Delete “4.5.15(b)/4.5.16(b)” and substitute “4.5.15/4.5.16” in three (3) places.

Special (a): Add “12 Ball to wall 100% 3.4 Gage”.

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4.4.2.48:

Delete note (b) in its entirety.

Delete note (g) in its entirety.

PAGE 73

Add new paragraph 4.4.3.1 as follows:

“4.4.3.1 Automated acceptance inspection equipment (AAIE). As part of the inspection equipment, it is the responsibility of the contractor to submit all design information and qualification test plan, which is applicable to the AAIE for Government approval. The contractor shall provide, for each defect associated with AAIE, manufacturing and inspection system which assures no more than one in a million fuzes contain the defect. This shall be assured by controlling the maximum defect rate produced and error rate of the inspection equipment such that the product of the two terms when multiplied together is less than one in a million.”

PAGE 75

DEFINITIONS:

ZONE A: Delete “ 1.4” + 1/64” ” and substitute “0.2500 ± 0.156 ”.

PAGE 78

4.5.12:

Line 6: Delete “less than 16 pounds” and substitute “less than 15.5 pounds”.

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4.5.15: Delete “Arming of the fuze is accomplished by raising of the arming pin, removal of the of the safety pin, and allowing the slider to move to its fully armed position. With the fuze in the armed state, an impulse shall be applied to the fuze/fixture in a direction parallel to the axis of the fuze within one degree.”

and substitute

“Arming of the fuze is accomplished by raising of the arming pin, removal of the of the safety pin, and allowing the slider to move to its fully armed position. Gage/measuring device shall be used to determine if the slider is in its fully armed position. If a total of 2 or more sliders fail to meet the fully armed requirement during the Go and NoGo tests, the lot shall be rejected. With the fuze in the armed state, an impulse shall be applied to the fuze/fixture in a direction parallel to the axis of the fuze within one degree.”

4.5.16: Delete “After mounting the fuze in the holding fixture, but prior performing the test, the arming pin shall be raised and the safety pin shall be removed. Removal of the safety pin shall be considered as beginning of the arming operation and thus the shielding operation applies.”

and substitute

“After mounting the fuze in the holding fixture, but prior performing the test, the arming pin shall be raised and the safety pin shall be removed. Gage/measuring device shall be used to determine if the slider is in its fully armed position. If a total of 2 or more sliders fail to meet the fully armed requirement during the Go and NoGo tests, the lot shall be rejected. Removal of the safety pin shall be considered as beginning of the arming operation and thus the shielding operation applies.”

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1340-1018)

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